

Supplementary Table 1. Discovery cohort genomic and clinical data.

Genomic features include: Ploidy, Purity, Heterogeneity, Genome with Anueploidy, BRAF_status, NRAS_status, # of mutations with snv multiplicity of 2 (n_snvmulti_2), number of mutation with snv multiplicity of 1 (n_snv_multi_1), ratio snv multiplicity, loss of heterozygosity (loh), snv_m_score (formula described in the paper), #clonal mutations, #subclonal mutations, total nonsynonymous mutations, Genome doubling and tumor mutational burden. Clinical features include: Best response, progressive disease, cohort, overall survival days, death, progression free survival days, progression, ECOG, Primary Type, Mstage, Brain metastasis, Lung metastasis, Liver metastasis, prior BRAF therapy.

Supplementary Table 2. Comparison of Heterogeneity measures and their predictive performance using discovery cohort (n=124).

Table showing the different measures of genomic Heterogeneity tested and the AUC achieved with the different models.

Supplementary Table 3. Previously described models and features tested.

Table of the models and features tested and the reference paper where the model was originally described.

Supplementary Table 4. Validation cohort.

Clinical and genomic features from the subgroup of patients ipilimumab naive at pre treatment with nivolumab or pembrolizumab from Freeman et al. (33) for which the RECIST response was available.

Supplementary Table 5. Cohort of patients treated with combo ICB (anti PD1+ anti CTLA4)

Table with clinical and genomic features of a cohort of metastatic melanoma patients treated with combination ICB.